Integrated Approach to Addressing Multiple SDG Challenges

Securing Safe Water and Implementing Agroforestry to Achieve Livelihood Improvement, Eradicate Child Labor, and Promote CO_2 Sequestration

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1. Introduction

1-1. About This Proposal

This proposal takes an integrated approach to complex issues such as global water problems, poverty, child labor, and CO_2 emissions.

- With less than five years remaining until the SDGs deadline (end of 2030), many challenges still remain.
- Many of these challenges are linked to the lack of safe, sanitary water.
- Reducing CO_2 concentrations may help address warming and mitigate climate change and natural disasters.

The factors behind the issues addressed by the SDGs are intricately interconnected.

Water & Sanitation

- Global water issues
- Contaminated water
- Water collection (lost education)

Society & Economy

- Poverty
- Child labor issues
- Unable to attend school

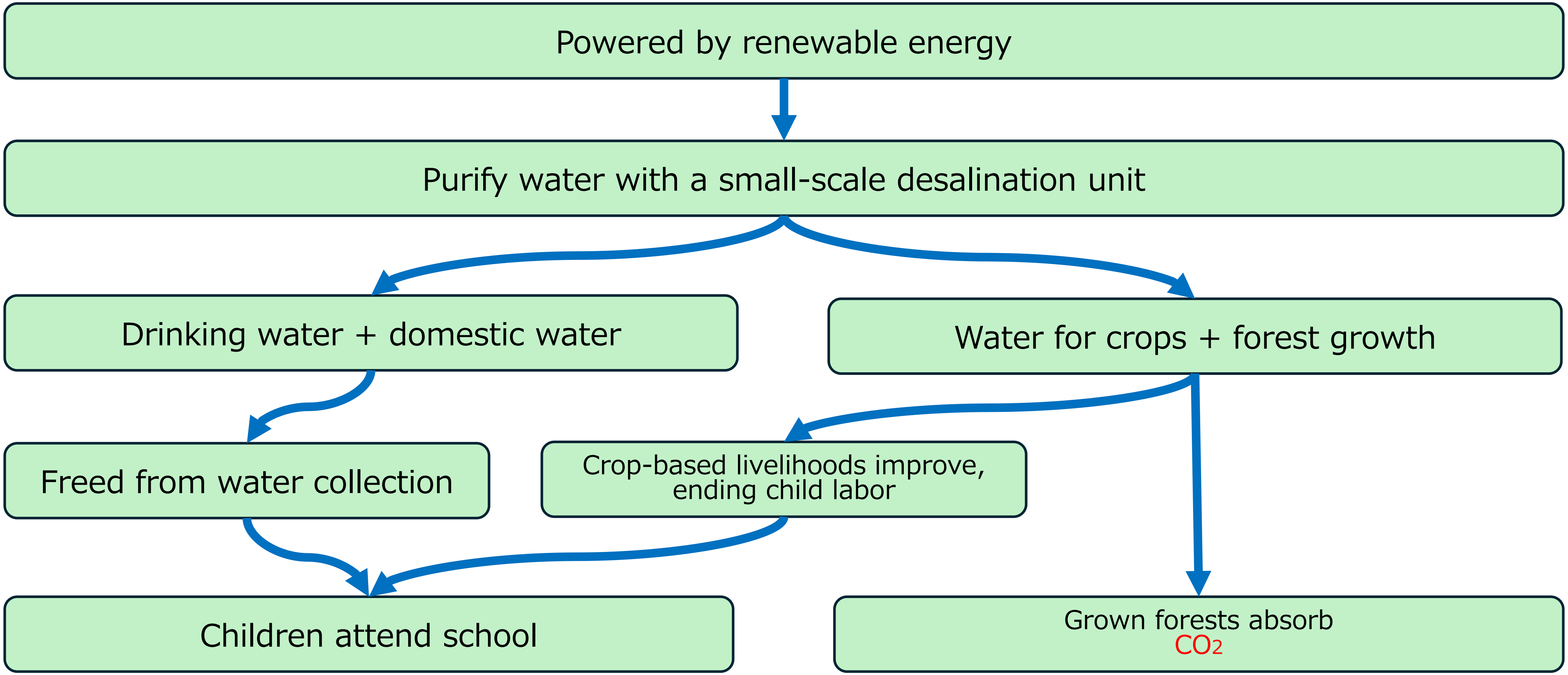
Agriculture & Environment

- Food shortages
- Climate vulnerability
- CO_2 reduction

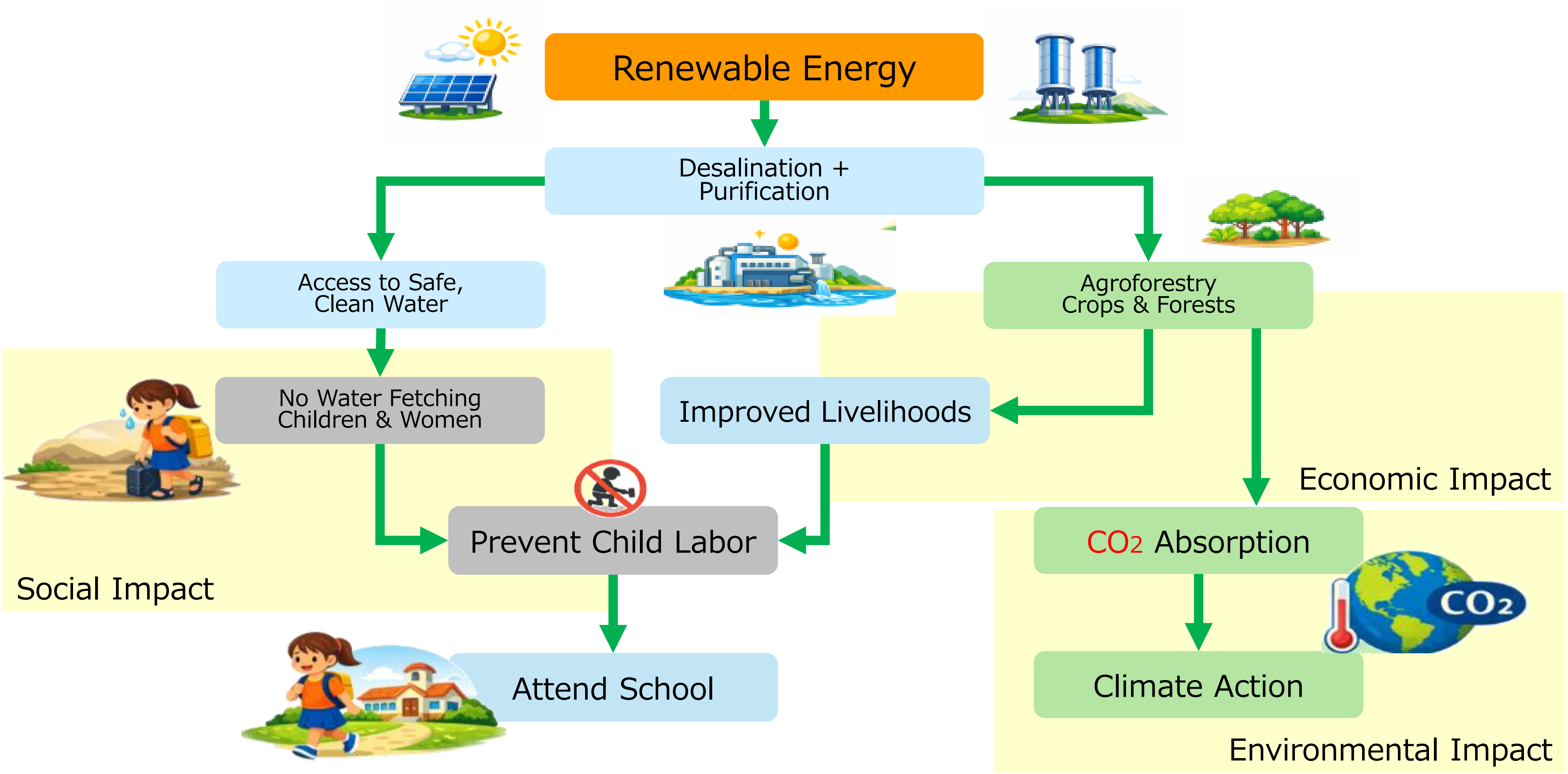
1-2. Key Technologies

Category	Technology	Impact Enabled
Infrastructure	Small-scale desalination unit Waqua MYZ-E250	Access to safe, sanitary water Freedom from water collection Water transport to inland areas
Energy	Renewable energy Silicon photovoltaics Perovskite photovoltaics Biodiesel	Power source for desalination Water transport to inland areas
Agriculture & Environment	Agroforestry	Livelihood improvement and local development through crops Prevent child labor and promote schooling CO_2 absorption and greening

1-2. SDGs: Overall View of a Simultaneous Approach to Multiple Issues



1-3. SDGs: Illustration of a Simultaneous Approach to Multiple Issues



1-4. Environmental Regeneration Effects of Agroforestry

- Climate Stabilization (Reducing CO_2 Concentrations)
  - Reduce CO_2 concentrations by 100–120 ppm, slowing warming.
  - Mitigate heat waves and natural disasters, restoring climate stability.

- Regeneration of Regional Ecosystems
  - Promote natural rainfall through increased forest cover.
  - Improve dryland climate adaptation by greening the Sahel region.

2. About Water

2-1. Water Requirements (Drinking Water + Domestic Water)

- International Standard (UNICEF)

  15 L/person/day　　　of which drinking water is 3–5 L

  https://www.unicef.or.jp/library/pres_bn2007/pres_07_33.html

- Assumption in This Model

  30 L/person/day　Amount assuming improved living standards.

- Water required for a settlement of 1,000 people

  30 L × 1,000 people = 30,000 L/day

2-2. Water Requirements (Crops + Forest Growth)

Water required for crops and forests

Required water per hectare for dryland agroforestry: **8,000–16,000 L/day/ha**
Based on CIFOR (Center for International Forestry Research) dryland studies
and technical documents such as “Agroforestry and Water Management in Drylands.”
Reference: CIFOR Publications (https://www.cifor.org/knowledge/publications/)
Considering evaporation on extremely hot days, salt accumulation prevention,
and climate-change risk (buffer), set the amount at about 1.7×, or **14,250 L/day/ha**.

When cultivated land is 20–40 ha, a large amount of water is required.
To avoid higher installation costs for small-scale desalination units,
adopt the water-saving technologies on the next page.

2-3. Water-Saving Technologies

Optimizing Water Retention for Crops and Forests

An approach to prevent excessive soil absorption and evaporation by “narrowing the outlet,” “covering the surface,” and “increasing water retention.”

- **Drip Irrigation**
- **Mulching (ground cover)**
- Use of water-retaining materials and soil conditioners
- Placement of windbreak forests (windbreaks)
- Subsurface irrigation (watering underground)

2-4. Water Requirements (1,000-person Settlement + 40 ha)

- Drinking & domestic water (1,000 people): 30,000 L/day
- Agroforestry water (40 ha): 570,000 L/day
  *Optimized by applying water-saving technologies (drip irrigation + mulching)

- Total daily water requirement 600,000 L/day
- Supply Plan
    - Water storage tank capacity: 900,000 L
      Always secure 1.5 days of supply; stable even during weather changes and maintenance
  - Transport system: Supply each inland area from desalination hubs using biodiesel water trucks

[Supplement: Required water by farmland area]

1 ha: 14,250 L / 20 ha: 285,000 L / 30 ha: 427,500 L / 40 ha: 570,000 L

3. About Seawater Desalination Systems

3-1. Small-Scale Seawater Desalination Unit

Purifies seawater and contaminated (raw) water.

- Desalination of seawater
- Purification by filtering river and contaminated water
- Purification using RO reverse-osmosis membranes
- Removes pathogens, viruses, and harmful substances
- Removes microplastics
- Secures tap-quality water
- Stable supply through distributed deployment due to compact size
- Track record in overseas water and sanitation projects

3-1. Small-Scale Seawater Desalination Unit

[Secondary Effects]

This unit uses RO reverse-osmosis membranes and removes fine particles in seawater with high efficiency.
In this process, it has been confirmed that more than 99% of microplastics (1–5 μm) can be removed.

[Future Development Potential]

By adding a dedicated module applying RO reverse-osmosis membrane technology, microplastic removal from seawater could be further enhanced.
This is promising as a new approach to addressing marine pollution.

4. About Renewable Energy

4-1. About Renewable Energy

- Silicon Photovoltaics
  - Conventional solar power technology
  - Based on the widely used 400 W panel.
  - Use salt-damage-resistant, corrosion-resistant panels

- Perovskite Photovoltaics
  - A solar power technology expected to be deployed in the future
  - Can generate power when applied to cylinders or glass surfaces
  - May surpass silicon photovoltaics in future cost competitiveness

5. Small-Scale Seawater Desalination and Required Number of Solar Panels

5-1. Required Units (MYZ-E250) and Solar Panels

	Seawater	Surface Water
Daily Water Demand	**600,000 L ***	**600,000 L ***
Solar Power Generation Hours	7:00-17:00(10h)	7:00-17:00(10h)
Water Production (10h/day)	2300 L/10hr	4600 L/10hr
Number of Units Required	Approx. 260 Units	Approx. 130 Units
Operating Voltage	Single-phase 200V	Single-phase 200V
Power Consumption	3,810 W(Unit+Pump) 3,360 W(Unit Only)	1,990 W(Unit+Pump) 1,540 W(Unit Only)
Required Solar Panels (400W) (Upper: Rated, Lower: 70% Efficiency)	Approx. 2,480 panels Approx. 3,540 panels	Approx. 1,240 panels Approx. 1,770 panels

* Slide 12: 2-4 require water(1000 people and 40 ha)

"260 units" is required due to the high pre-treatment load for seawater.
Figures are based on the current model (MYZ E-250).
Appendix cost estimates adopt the 70% efficiency figures (Seawater: 3,540, Surface Water: 1,770).

5-2. Selection of Water Intake Sites and Water Supply

- Selection Criteria for Water Intake Sites
  - Physical requirements: Calm sea (seawater desalination) or rivers, lakes, aquifers
  - Sustainability: Stable intake volume throughout the year (minimum: 600 m^3/day)
  - Social requirements: Sufficient consensus-building with local residents

- Delivery & Logistics (areas where securing water sources is difficult)
  - Securing delivery methods:
    - Current: Biodiesel water trucks
    - Future: EV water trucks equipped with sodium-ion batteries

6. Agroforestry: Crops and Trees

6-1. Agroforestry Structure and Value

Perspective	Role of Agroforestry	Economic & Environmental Value Created
Climate Resilience	Trees shade intense sunlight and moderate microclimates	Prevents crop die-off during droughts (risk diversification)
Soil Environment	Leaf litter and roots add organic matter and nurture soil	Reduces chemical fertilizer dependence and improves long-term soil fertility
Revenue Structure	Multi-crop, multi-layer cultivation diversifies harvest timing	Stabilizes income year-round and breaks the poverty cycle *1
Regional Development	Adoption of the method triggers local ecosystem regeneration	Improves working conditions and builds sustainable communities

*1 Helps reduce child labor tied to household income.

6-2. Combination of Locally Suitable Crops and Trees

■ Crops/Trees to Grow and Their Roles

Category	Role	Key Examples
Livelihood Improvement	Cash income from harvest and sales	Sweet potato, okra, cassava, vanilla, cacao, mango, papaya, avocado
Environmental Restoration	Soil improvement & CO_2 absorption (carbon fixation)	Moringa, acacia, teak
Fuel	Fuel for biodiesel water trucks	Moringa, jatropha, sunflower, castor bean, rapeseed

Note: Crop and tree selection should respect local food culture and expert guidance.

6-3. Global Environmental Contribution of 10,000 Sites

40 ha × 10,000 sites delivers approx. 6 million tons of annual CO_2 absorption

Comparable to the CO_2 emissions of Latvia, Iceland, and Luxembourg.

*Model case; results vary by vegetation density and climate conditions.

Average estimates below use a 1,000-person settlement model (20–40 ha), since carbon sequestration varies by crop/tree type.

Cultivated Area (per settlement)	Annual CO2 Absorption (per site)	Number of Projects	Total Annual CO2 Absorption
20 ha	200–400 tons	×10,000	2–4 million tons
30 ha	300–400 tons	×10,000	3–4 million tons
40 ha	400–800 tons	×10,000	4–8 million tons

6 million tons is an average estimate.

7. Application Sites

7-1. Target Settlements Requiring Deployment of This Integrated Model

- Africa: 54 countries

  Africa has a population of 1.4 billion; over 400 million people are said to lack access to safe water.

- African settlements

  300–1,500 people (most common)

  2,000–5,000 people (medium-sized)

  10,000+ people (treated as towns)

- Settlements with water issues

  Hundreds of thousands of settlements in Africa face water-related issues.

  Depending on the country, thousands to tens of thousands of settlements may be target sites.

- Estimated applicable areas

  Estimated at around 500,000 settlements.

7-2. Target Regions Requiring Deployment of This Integrated Model

- Applicable to Southeast Asian regions with water issues
  Bangladesh, Cambodia, Laos, Myanmar, parts of Indonesia, parts of the Philippines

- Applicable to other regions as well
  Desert and semi-arid regions across the Northern Hemisphere
  Potentially deployable in ways that consider water resources and ecosystems.

  Northern China
  Around Mongolia
  Central Asia (southern Kazakhstan, Uzbekistan, Turkmenistan)
  Middle East (Jordan, northern Saudi Arabia, western Iraq)

8. Other Considerations

8-1. Other Considerations: Do Not Harm

- **Must not become a source of conflict**

  Facilities built by this project must not become a flashpoint between the national government and anti-government armed groups.

  Local residents' understanding is needed to avoid any sense of unfairness between project areas and non-project areas.

8-2. Other Considerations: Implementation Sites

- **Implementation sites (solar panels, small-scale seawater desalination, agroforestry)**

  Land should be secured as state-owned land or as use rights agreed with the community,

  so the design avoids land grabbing and inequity between regions.

- **Water allocation**

  How to decide allocation of drinking/domestic water, agricultural water, and water for growing forests.

- **Salt production**

  Concentrated brine discharged from small desalination/water purification units (when using seawater) can be used to produce salt.

  Though limited, it can help improve local livelihoods and may cover maintenance costs such

  as filters.

8-3. Other Considerations: IT-Related

■ Lithium-ion batteries: do not use fossil fuels

Adopt cobalt-free LFP (lithium iron phosphate) batteries to reduce conflict-mineral and child-labor risks
while maintaining safety in high-temperature environments.

Note: Lithium-ion batteries may contain conflict minerals such as cobalt, and harsh child labor at mining sites is a concern.

From a CO_2 reduction perspective, do not use generators powered by fossil fuels.
If desalination must operate after sunset, use electricity from biodiesel generators.
Consider reducing the number of units; until large **sodium-ion batteries** become commercialized,
purify as much water as possible during daylight hours and store it in tanks.

Note: Biodiesel is a renewable fuel, but its lifecycle greenhouse-gas emissions depend
on the feedstock, production process, transport, and land-use impacts.

8-4. Other Considerations: Operations

- **Build an IT-based remote monitoring system**

  Monitor power generation, detect desalination/purification unit anomalies, and alert for filter replacement.

  Because the units do not support management protocols, monitor them with remote surveillance cameras.

  Note: Saving camera footage and temporarily storing logs will be designed during implementation.

- **Equipment maintenance**

  Regular maintenance is required to operate desalination/purification units, solar power systems, and wind power systems.

  Skill transfer to local residents is necessary.

- **Lightning rods**

  Always install lightning rods to prevent equipment failure caused by lightning strikes.

8-5. Other Considerations — Risk Avoidance: Desert Locust Countermeasures (Short / Medium / Long Term)

- Risks posed by desert locusts
  - Can devastate crops and forests in a short time
  - Gregarious swarms range from tens of millions to billions of individuals
  - Travel more than 100 km per day
  - The early stage of agroforestry is the most vulnerable

- Short term: Leverage FAO's early warning system
  - FAO Locust Watch: continuously monitor outbreak status, migration forecasts and alert levels.

- Medium term: Eliminate at the larval (hopper) stage (carried out at village level)
  - Larvae (hoppers) have no wings and only walk on the ground — this stage is the easiest to control
  - Methods: herd into trenches and bury, barrier traps, small-scale spraying (conventional spraying, not ULV)
  - Manageable at the village level → the most practical way to prevent major outbreaks

- Long term
  - Increasing forest cover
  - Natural predators (birds, insects, small animals)
  - Soil stabilizes and desertification is curbed → over the long term this reduces the very breeding grounds of desert locusts

9. What to Expect in the Future

9-1. Increase in Vegetation Through Natural Rainfall

By deploying this model across multiple sites in suitable regions, increased vegetation and evapotranspiration could potentially contribute to changes in local water-cycle dynamics in some arid areas.

Under favorable regional climate and atmospheric conditions, changes in water-cycle dynamics may contribute to more stable rainfall patterns. In addition to planted forests, naturally regenerated vegetation could spread, potentially increasing CO_2 sequestration.

This could potentially contribute to a positive feedback loop that supports vegetation recovery and may help reduce desertification in some arid regions.

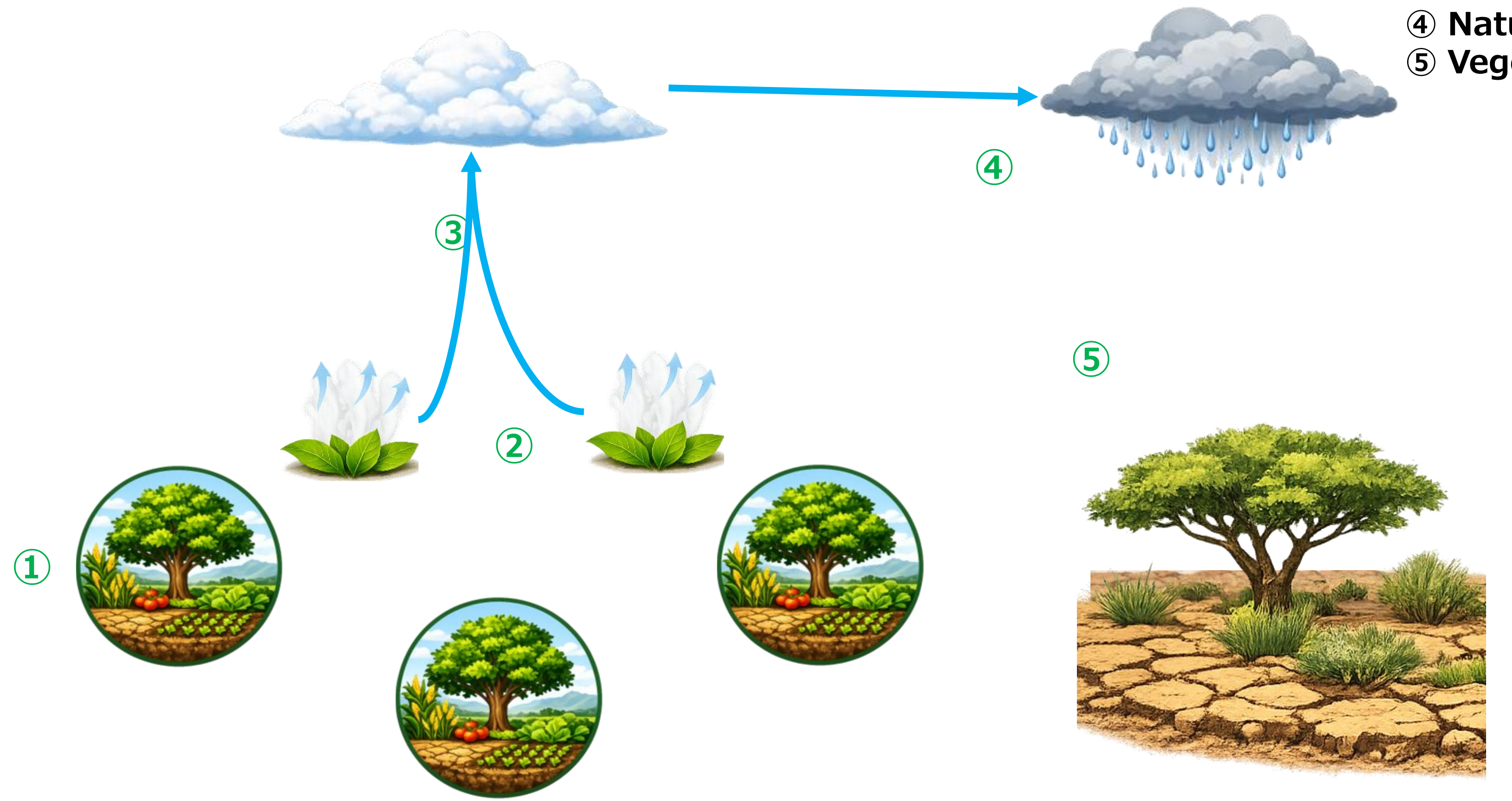
The specific implementation regions and forest areas should be determined by experts, but

for example, in an arid area about the size of Narita Airport (approx. 1,000 ha), restoring about 100–300 ha as forest

could have a meaningful positive effect on natural rainfall and water-cycle recovery in the surrounding area.

9-2. Natural Rainfall Mechanism — Conceptual Image

① Deploy multiple agroforestry sites
② Transpiration from forests
③ Increased evapotranspiration
→ atmospheric moisture and
cloud formation
④ Natural rainfall
⑤ Vegetation establishes naturally



9-3. Crop Production + Forest Growth with Other Farming Methods

Once a certain level of natural rainfall can be expected, combining the following farming methods can make crop production possible even in arid areas.

- **Half-moon farming**

  → Rainwater collects effectively in the basins

- **Zai farming**

  → Water and nutrients gather in the holes, sharply increasing germination rates

- **Stone lines**

  → Water flow slows, helping soil remain instead of being washed away

- **FMNR (Farmer Managed Natural Regeneration)**

  → A core technique for forest restoration, making forest regeneration easier

9-4. Reducing Hunger and Conflict

Through this model, water, agriculture, and livelihoods can become stable, enabling many regions to prosper and **reduce hunger**.

Disputes over local resources and migration for livelihoods may decrease, potentially reducing the risk of “competition-driven conflict.”

This not only satisfies the basic principle of international cooperation,

“Do No Harm (avoid causing harm)” but could also contribute to regional stability and peacebuilding.

10. Summary

10-1. Summary: A Three-Fold Impact(1/3)

Three Core Transformations

1.Water Security:

Providing stable access to safe water, eliminating the burden of water collection, and enabling educational opportunities.

2.Agroforestry & Ecosystem Regeneration:

Driving economic growth, helping to reduce child labor associated with household poverty,

and creating infrastructure designed to minimize reliance on fossil fuels and powered primarily by renewable energy.

3.Sustainable Community Building:

Creating fossil-fuel-free infrastructure powered by renewable energy.

10-1. Summary: A Three-Fold Impact(2/3)

Standardization of the 1000-Person Unit Model

- **Unified Design:**

  Standardized capacity, equipment, and renewable energy configuration for any region.

- **Versatile Application:**

  Fully compatible with both seawater and freshwater sources.

- **High Scalability:**

  Designed for universal implementation, ensuring ease of deployment across diverse countries.

10-1. Summary: A Three-Fold Impact(3/3)

Scaling Potential & Vision

■ **Broad Reach:**

Applicable to hundreds of thousands of communities across Africa and Asia.

■ **Environmental Impact:**

Achieving 6 million tons of annual CO_2 reduction through 10,000 units.

■ **Social Impact:**

Improving lives for hundreds of millions through simultaneous water and education reform.

■ **Global Contribution:**

Establishing a Japanese-led international model to solve global challenges.

"Targeting Broad Deployment by 2040"

10-2. Relationship with the SDGs

1 NO POVERTY	Improving livelihoods through crops Escaping poverty through regional development
2 ZERO HUNGER	Securing food through crop harvests
4 QUALITY EDUCATION	Stop fetching water Higher incomes reduce child labor and enable school attendance
5 GENDER EQUALITY	Relieve women and girls of water-carrying duties Education → better jobs → prevention of child marriage
6 CLEAN WATER AND SANITATION	Securing safe, hygienic water Safe toilets with electric lighting
7 AFFORDABLE AND CLEAN ENERGY	Use of renewable energy
13 CLIMATE ACTION	CO_2 absorption (tree planting, agroforestry, etc.)
15 LIFE ON LAND	Forest growth Crops through drip irrigation and mulching

Supports multiple SDGs

This proposal was created for **children of the future**.
I want to reduce the number of children who must fetch water,
work to support their families,and give up school.

I want to reduce atmospheric CO_2 and the number of
children suffering from heat waves and droughts.
I want to create a future where, as adults, they can help build hope
for the next generation.
That is the spirit behind this proposal.

Thank you for your consideration.

Open Knowledge Statement

I am entrusting this to a promising young person.

22 Apr 2026 (First edition)

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This proposal receives technical cooperation from Waqua Co., Ltd. for compact seawater desalination equipment (MYZ-E series).

*This document does not cover detailed specifications related to corporate confidential information.

■ Revision History

Date	Changes	Notes
22 Apr 2026	First edition published	
25 Apr 2026	Updated to Rev. 1.1; added information on microplastic removal	
18 Jul 2026	Updated to Rev. 1.2; revised title and content. Updated the Japanese version and released the English version.	